

DMS and ADAS Parameter Setting User Manual

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Contents

1	Overview	3
2	First-time configuration	6
3	DMS configuration in each AI feature	7
3.1	Fatigue	7
3.1.1	Sensitivity recommendation according to various scenarios	8
3.2	Distraction	9
3.2.1	Sensitivity recommendation according to various scenarios	11
3.3	Yawn	11
3.4	Handheld Devices	12
3.5	Smoking	12
3.6	No Seatbelt	13
3.7	No Driver	14
3.8	Camera Covering	15
3.9	Infrared block alarm	15
3.10	Eat& Drink	15
3.11	No Mask	16
4	ADAS	17
4.1	Lane departure warning	17
4.1.1	Boundary scenarios	17
4.2	Forward collision warning	18
4.2.1	Boundary scenarios	18
4.3	Headway monitoring warning	19
4.4	Pedestrian collision warning	20
4.4.1	Boundary scenarios	20
4.5	Speed limit warning	21
4.6	Rolling stop	22
5	Different trucking industry customer recommended settings	23
5.1	Long-Haul Trucking	23
5.2	Medium-Short Haul Freight	24
5.3	Urban Delivery Services	24
5.4	Hazardous Materials Transportation	24

1 Overview

In the actual application of the Driver Monitoring System (DMS) and Advanced Driver Assistance System(ADAS) , the computer vision of artificial intelligence has its boundary scenarios. For instance, when the human eye cannot see the seatbelt, the AI cannot detect it either. However, we have found that many false positives and missed detections in DMS often stem from improper parameter settings. Additionally, the needs for a DMS system vary across different trucking companies, transportation scenarios, and drivers' characteristics. While standardized settings can meet basic requirements, they often fail to achieve optimal performance. Personalized parameter configurations allow the DMS system to better adapt to specific usage environments, maximizing its value. Therefore, it is crucial to understand the operational mechanisms of each parameter in order to optimize system performance and user experience.

This document provides a detailed explanation of the parameter configuration methods for the DMS and ADAS system's various functions. Chapter 1 offers a detailed introduction to the meanings of parameters in each configuration interface; Chapter 2 outlines the initial setup process in detail; Chapter 3 and 4 demonstrate effects of various sensitivity settings based on the alarm functions of each AI feature, respectively; In Chapter 5, professional configuration recommendations for sensitivity are provided, tailored to different types of trucking operations, helping them fully leverage the safety assurance role of the DMS system; common issues and possible solutions will be discussed in the final Chapter 6.

Figure 1 demonstrates an example of a parameter configuration interface with fatigue detection.

The image shows a dark-themed configuration window titled "Driver Fatigue Trigger". It contains several settings:

- Lvl1 Speed Range:** Two input boxes with values "20" and "50", a tilde "~" between them, and the unit "km/h".
- Lvl2 Speed Range:** A label "≥" followed by "50 km/h".
- Sensitivity:** A dropdown menu currently showing "High".
- Effective Time:** A label followed by a dropdown menu showing "Low", "Middle", "High", and "User-Defined". To the right of the dropdown is the text "~ 600)Seconds".
- Do not refresh the alarm timer:** A checkbox that is currently unchecked.
- Buttons:** "Cancel" and "OK" buttons at the bottom.

Figure 1: An example of a parameter configuration interface

1. Lvl1 Speed Range

Range: Different alarm types have different default configurations, and this parameter can be customizable by the customer.

This is the effective speed range for triggering a Level 1 alarm. When the vehicle's speed falls within this range, the system will detect and trigger a Level 1 alarm based on

Table 1.1: Sensitivity parameter settings for each AI alert (Unit: s)

Alarm	High Sensitivity	Medium Sensitivity	Low Sensitivity	User-defined
Fatigue	Table 3.1 and Table 3.2			
Distraction(L+R)	4	10	14	1-60
Distraction(U+D)	2	3	4	1-60
Yawn	1	2	3	1-60
Handheld Device	4	6	12	1-60
Seatbelt	10	20	30	1-60
No driver	5	15	20	1-60
Camera cover	5	15	20	1-60
Smoking	2	3	5	1-60
No mask	3	10	20	1-60
Eat	4	6	8	/
Drink	1	2	3	/
Infrared Block	2	5	10	1-60

other conditions, with voice and beep reminders of R-Watch.

2. Lvl2 Speed Range

The speed range for triggering a Level 2 alarm starts from the maximum speed of the Level 1 alarm.

When the vehicle's speed exceeds Level 1 speed, the system will trigger a Level 2 alarm based on certain conditions. The threshold for the Level 2 alarm depends on the configuration of the Level 1 alarm. Only a beep sound reminder will be provided on the device side.

3. Sensitivity

Table 1.1 provides the alarm times for the high, medium, and low sensitivity settings of various DMS-related functions.

Sensitivity determines the strictness of DMS detection, measured in seconds.

4. Effective Time

Value: customizable within 600 seconds.

Meaning: When an alarm is triggered, if the same type of alarm is triggered again within the set "Effective Time" range (e.g., 60 seconds), the system considers these alarms as part of the same event (uploaded to the online platform), rather than a new alarm. The Effective Time controls the merging of repeated alarms within this duration, reducing redundant records and evidence uploads to the platform while still maintaining the R-watch reminder function in the driving cockpit.

5. Do not refresh the alarm timer

- If not enabled (refresh the alarm timer): each time a new alarm of the same type is triggered within the Effective Time (0–600 seconds, customizable), the timer resets to start counting from the end moment of the new trigger. This means the effective time is refreshed and follows behind each subsequent trigger.

- If enabled (do not refresh the alarm timer): the monitoring of the customized "Effective Time" (set between 0–600 seconds) will be fixed. This means that once an alarm is triggered, the effective time period starts counting from that moment and will not be refreshed by subsequent triggers of the same alarm type within that period.



Figure 2: Difference between "Refresh the alarm timer" and "Do not refresh the alarm timer" (Alarm 1, Alarm 2, and Alarm 3 are the same type of alarms.).

2 First-time configuration

For first-time users of the DMS, it is recommended to follow the steps below for initial configuration to ensure the system performs optimally in your specific usage environment.

Step 1: Assess Usage Scenarios

Before making any settings, customers should carefully evaluate their actual usage scenarios. This includes analyzing vehicle types, management needs, primary driving routes, driver experience levels, and operational time schedules. Different usage scenarios have significantly varying requirements for DMS, and only by accurately understanding your needs can you select the most suitable parameter configuration.

Step 2: Configure Speed Range Parameters

The speed range settings should be adjusted based on your actual driving environment.

Attention: if the actual driving speed falls outside these two configured speed ranges, the DMS will not generate any alerts.

Step 3: Select Base Sensitivity Level

Based on the results of the scenario assessment, choose an appropriate base sensitivity level. It is recommended that new customers start with the medium sensitivity level, which provides a balanced setting that offers basic safety assurance while minimizing excessive interruptions.

Step 4: Set Effective Time Parameters

The setting of effective time parameters should consider your operational mode and usage scenario. For long-haul transportation or scenarios requiring continuous monitoring, it is recommended to set the effective time to 0-100 seconds, enabling a continuous monitoring mode to ensure driver status is tracked throughout the journey. For short-haul transportation or intermittent work scenarios, an appropriate effective time, such as 300–600 seconds, can be set. This balances safety with user experience and reduces data usage.

Step 5: Test and Adjust

Week 1: After completing the initial setup, it is recommended to conduct a one-week test run. During this period, record detailed information about system alerts, including alert times, reasons for alerts, and the driver's actual state at the time.

Week 2: Based on the data feedback from the first week, adjust the sensitivity downward for functions with a high false alarm rate and increase the sensitivity for functions with suboptimal detection performance.

Week 3: Perform fine-tuned adjustments, tailoring settings to the characteristics of different drivers and routes.

Week 4: Establish a continuous optimization mechanism, regularly analyzing system data to refine and improve parameter settings.

3 DMS configuration in each AI feature

3.1 Fatigue

IPC Cameras (e.g., C29N, AD Kit3.0) and AHD Cameras (e.g., CA29M) employ distinct algorithm logic for fatigue detection. Table 3.1 presents the algorithm logic for IPC cameras, while Table 3.2 details the logic for AHD cameras. A comparison of sensitivity levels is provided in Table 3.3, with additional insights into various scenarios illustrated in the following.

IPC camera fatigue algorithm logic

The fatigue detection algorithm of Streamax[®] is powered by a comprehensive, high-quality training dataset derived from real-world collections. This dataset, encompassing over 1000,000 annotated video clips, ensures robust model generalization across diverse scenarios, including varying lighting, head poses, and driver demographics.

Streamax’s algorithms integrate deep learning-based real fatigue detection algorithm, targeting subtle behaviors to provide real-time alerts, it does not rely solely on predefined rules such as eye closure duration or blink frequency to determine fatigue. Instead, it is trained on millions of real fatigue video clips using a deep neural network model, learning fatigue features from multidimensional cues including gaze stagnation, eye closure, blinking, micro-expressions, and head orientation to comprehensively assess the driver’s fatigue state. The whole processing time on a Neural Processing Unit (NPU) only takes 3-4 milliseconds on IPC camera. As a result, C29N will compute a deep learning probability according to the driver’s behavior.

IPC fatigue logic detects fatigue based on deep learning probability and cumulative eye closure durations within a 3-second window. Eye closures are considered if they last over 200ms. The fatigue alert will be triggered if one of the following conditions is met.

Sensitivity	Condition 1	Condition 2
High	Once deep learning fatigue detection instantaneous probability > 0.9, AND cumulative eye closures (> 200ms) in 3s window > 500ms.	Cumulative eye closures (> 200ms) in 10s window > 2s.
Medium	Once deep learning fatigue detection instantaneous probability > 0.95, AND cumulative eye closures (> 200ms) in 3s window > 800ms.	Cumulative eye closures (> 200ms) in 10s window > 3s.
Low	Once deep learning fatigue detection instantaneous probability > 0.95, AND cumulative eye closures (> 200ms) in 3s window > 1000ms.	Cumulative eye closures (> 200ms) in 10s window > 4s.
Custom	Once deep learning fatigue detection instantaneous probability a <i>value%</i> (customized range: 75%–95%), AND cumulative eye closures (> 200ms) > 800ms(Fixed).	Cumulative eye closures (> $X \times 100$ ms) in 10s window > N seconds.

Table 3.1: IPC camera fatigue detection logic

AHD camera fatigue algorithm logic

This logic detects fatigue based on continuous eye closure durations and cumulative eye closures within a 10-second window. Eye closures are considered if they last over 200ms in the cumulative condition. The fatigue alert will be triggered if one of the following conditions in Table 3.2 is met.

Sensitivity	Condition 1	Condition 2
High	Continuous eye closure for 2 seconds.	Cumulative eye closures ($> 200\text{ms}$) in 10s window $> 2\text{s}$.
Medium	Continuous eye closure for 3 seconds.	Cumulative eye closures ($> 200\text{ms}$) in 10s window $> 3\text{s}$.
Low	Continuous eye closure for 4 seconds.	Cumulative eye closures ($> 200\text{ms}$) in 10s window $> 4\text{s}$.
Custom	Continuous eye closure for N seconds.	Cumulative eye closures ($> X \times 100\text{ms}$) in 10s window $> N$ seconds.

Table 3.2: AHD camera fatigue detection logic

Table 3.3: Comparison analysis of fatigue sensitivity settings

Sensitivity	False negative	False positive	Timeliness	User experience
High	Very Low	High	Excellent	Moderate
Medium	Low	Moderate	Good	Good
Low	High	Low	Moderate	Excellent

3.1.1 Sensitivity recommendation according to various scenarios

Night Long-Haul • Recommended: **High or Medium**

- High Sensitivity: excellent timeliness, high false positives (e.g., IPC cameras, probability > 0.9 & $> 500\text{ms}$ closures or 2s closures)
- Medium Sensitivity: good timeliness (e.g., IPC cameras, probability > 0.95 & $> 800\text{ms}$ closures or 3s closures)
- Low Sensitivity: moderate timeliness, possible false negatives (e.g., IPC cameras, probability > 0.95 & $> 1000\text{ms}$ closures or 4s closures)

Urban Congestion • Recommended: **Low or Medium**

- High Sensitivity: excellent timeliness, high false positives
- Medium Sensitivity: good timeliness, moderate false positives
- Low Sensitivity: moderate timeliness, low false positives

Highway • Recommended: **High or Medium**

- High Sensitivity: maximum safety, very low false negatives
- Medium Sensitivity: balanced safety, low false negatives

- Low Sensitivity: fair safety, high false negatives

Short Delivery • Recommended: [Low or Medium](#)

- High Sensitivity: over-sensitive, moderate user experience
- Medium Sensitivity: good user experience, moderate detection
- Low Sensitivity: excellent user experience, non-intrusive

Hazmat Transport • Recommended: [High or Custom mode](#)

- High Sensitivity: top safety, very low false negatives
- Medium Sensitivity: high safety, low false negatives
- Low Sensitivity: low safety

Novice Driver • Recommended: [High or Medium](#)

- High Sensitivity: full protection, very low false negatives
- Medium Sensitivity: moderate protection, low false negatives
- Low Sensitivity: less protection, high false negatives

3.2 Distraction

Figure 3: Configuration of distraction trigger

- **Sensitivity Settings:** Options include Low, Medium, High, and User-Defined, allowing customization of the system's responsiveness to detect driver behavior. The durations of sensitivities for distraction are shown in [Table 1.1](#).

– [How does the sensitivity time work in the distraction detection?](#)

Streamax[®] designs its distraction logic aligning international standards such as Euro NCAP and leading Tier-1 practices in designing the logic of distraction alarms.

The DMS triggers a distraction alarm the instant any sliding sensitivity time window contains over 80% accumulated time with the driver's eyes off the forward roadway. The logic is shown in [Figure 4](#).

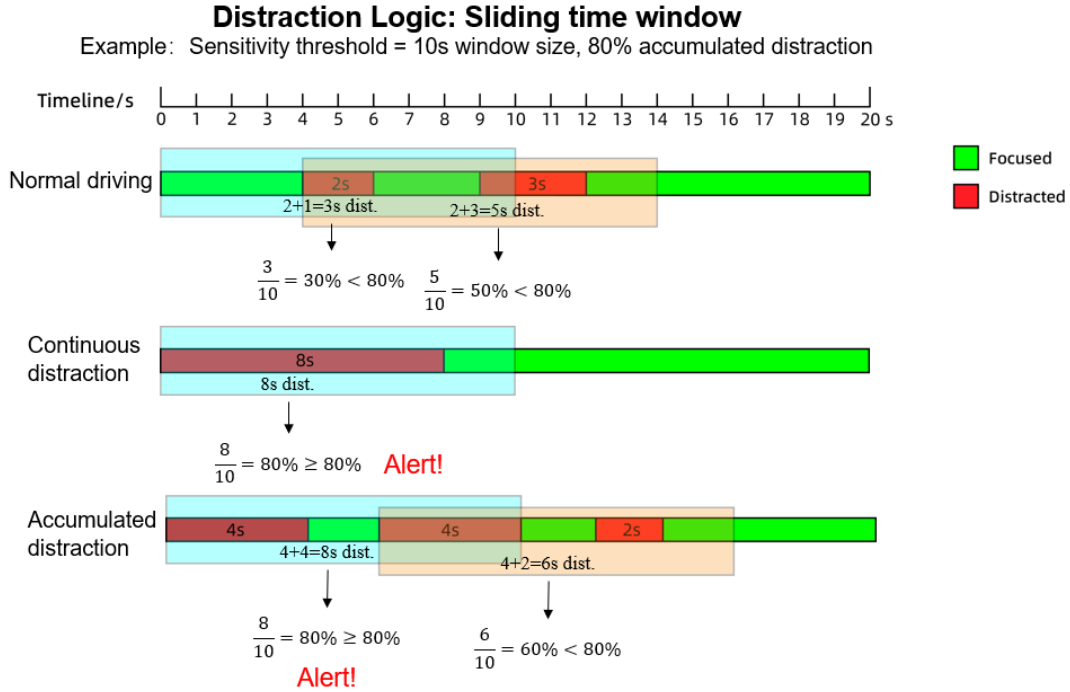


Figure 4: Examples of distraction logic

- **Lane Departure Suppression:** When enabled, this feature suppresses distraction alerts during intentional lane changes (e.g., turning right or left), preventing unnecessary warnings.
- **Judgment Modes:** Available options are L+R (Left + Right), Up+Down, and L+R+Up+Down. The lane departure suppression setting is displayed only when Left and Right side distractions are being evaluated.



Figure 5: Distraction levels according to deflection angles of a driver's head

- **Distraction Level (Deflection Angle):** Shown in the Figure 5 and defined as follows:
 - light: Approximately 24 degrees
 - medium: 28 degrees
 - high: Greater than 32 degrees

3.2.1 Sensitivity recommendation according to various scenarios

Night Long-Haul Transportation • Recommended: High or Medium

- High Sensitivity: Overly strict, may interfere with necessary observations like rearview mirrors (e.g., left/right distraction for 4s or up/down distraction for 2s)
- Medium Sensitivity: Balances timeliness and false alarms, suitable for monotonous road conditions (e.g., left/right distraction for 10s or up/down distraction for 3s)
- Low Sensitivity: Moderate timeliness, may miss alerts (e.g., left/right distraction for 14s or up/down distraction for 4s)

Urban Congestion • Recommended: Medium or Low

- High Sensitivity: High false alarms, frequent observation required in urban settings
- Medium Sensitivity: Moderate detection, allows flexibility
- Low Sensitivity: Moderate timeliness, low false alarms

Highway • Recommended: Medium

- High Sensitivity: High safety, but may falsely alert during navigation checks
- Medium Sensitivity: Balances safety, low missed alerts
- Low Sensitivity: General safety, suitable for experienced drivers

Short-Haul Delivery • Recommended: Medium or Low

- High Sensitivity: Interferes with frequent turns and observations
- Medium Sensitivity: Good user experience, moderate detection
- Low Sensitivity: Excellent user experience, non-intrusive

Hazardous Materials Transportation • Recommended: High

- High Sensitivity: Requires maximum attention
- Medium Sensitivity: High safety, low missed alerts
- Low Sensitivity: Not recommended, high risk

Novice Driver • Recommended: High or Medium

- High Sensitivity: Comprehensive protection
- Medium Sensitivity: Moderate protection
- Low Sensitivity: Less protection, high missed alerts

3.3 Yawn

The yawning alert is triggered when the system detects that the mouth's opening amplitude, as shown in Figure 6, persists for a set sensitivity time (High, Medium, Low and User-Defined are shown in Table 1.1).

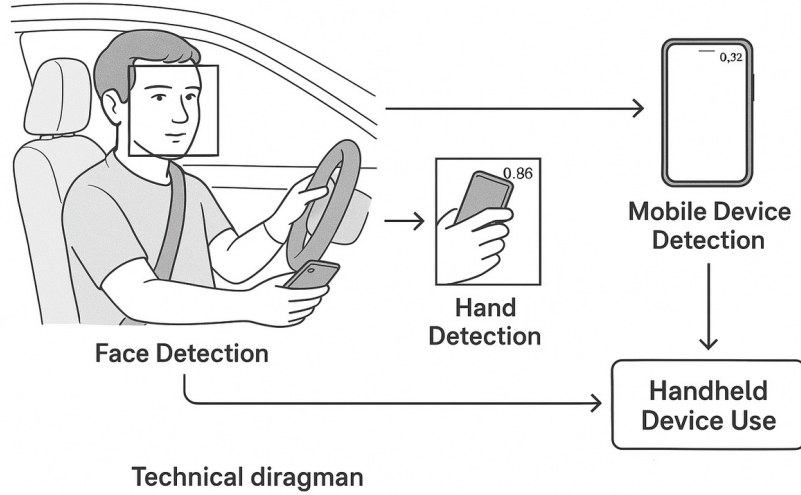


Figure 8: Flowchart of phone used detection



Figure 6: Yawning



Figure 7: Yawning with a hand covering

Boundary scenarios: When the driver is yawning with a hand covering his/her mouth as shown like Figure 7, the yawning alert will not be triggered since DMS can't detect driver's mouth.

3.4 Handheld Devices

When a face is detected, the system performs a combined recognition of mobile device features and gesture characteristics as shown in Figure 8. If both a hand and a mobile device are identified, DMS classifies this as a "handheld device use" state. If this state persists beyond the set sensitivity time threshold (e.g., 0–60 seconds in User-Defined mode, more details are listed in Table 1.1), an alarm will be triggered.

3.5 Smoking

When a face is detected, the system performs a combined recognition to identify cigarette presence. If a cigarette is detected in the mouth area, the system classifies this as a

”smoking” state. If this state persists beyond the set sensitivity time threshold as shown in Table 1.1, an alarm will be triggered.

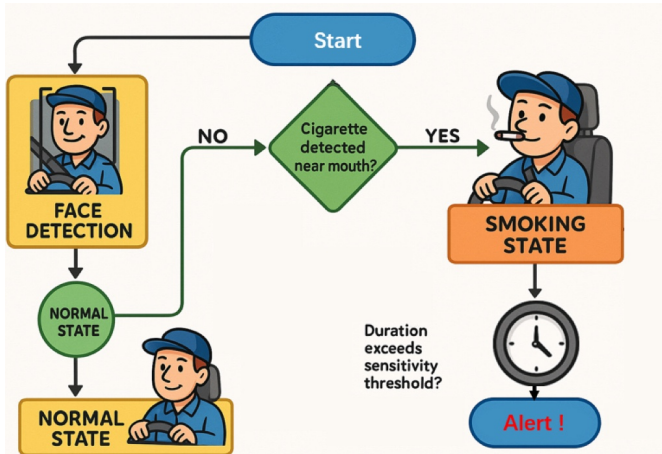


Figure 9: Flowchart of smoking detection

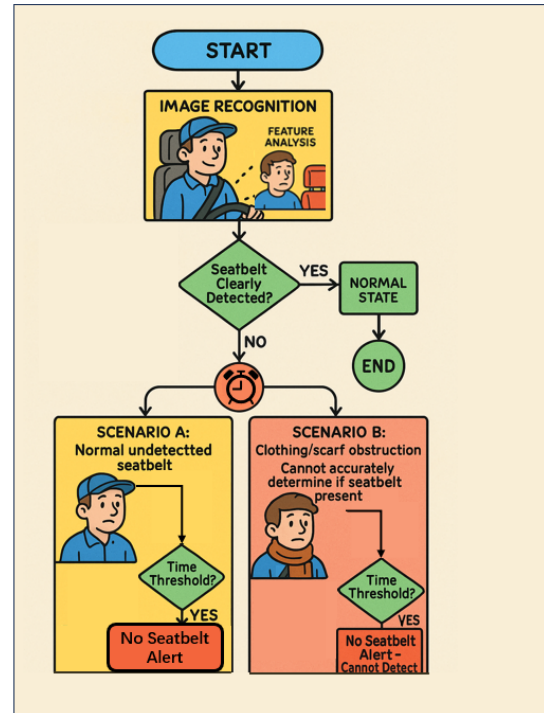


Figure 10: Flowchart of seatbelt detection

3.6 No Seatbelt

Seatbelt Trigger

Alarm mode: Normal mode ▾

Lvl1 Speed Range: 30 ~ 50 km/h

Lvl2 Speed Range: ≥ 50 km/h

Sensitivity: User-Defined ▾ 60 (0 ~ 60)Seconds

Effective Time: 0 (0 ~ 600)Seconds ?

Do not refresh the alarm timer: ☐ ?

Copilot detection enable: ☐ ?

Cancel OK

Figure 11: Normal mode of seatbelt detection

Seatbelt Trigger

Alarm mode: Regular inspection ▾

Lvl1 Speed Range: 30 ~ 50 km/h

Lvl2 Speed Range: ≥ 50 km/h

Sensitivity: User-Defined ▾ 60 (0 ~ 60)Seconds

Copilot detection enable: ☐ ?

Inspection interval: 60 (5 ~ 240)Minute

Cancel OK

Figure 12: Regular inspection mode of seatbelt detection

The system conducts image recognition feature analysis (Figure 10) of the driver or co-pilot’s body area, and co-pilot detection will take effect only when it is enabled. If a seatbelt is not detected and this state persists beyond the set sensitivity time threshold,

as shown in Table 1.1, an alarm will be triggered.

Boundary scenarios: The system will report a seatbelt alert if it is mostly (around 80%) obstructed. These kinds of false positives can probably be solved by integrating with DSC (supported by AD Plus 2.0).

Seatbelt detection offers 2 modes:

- Normal mode (default): the configuration shown in Figure 11
- Regular inspection mode: as shown in Figure 12, the seatbelt is detected only once per "Inspection interval". For example, if the "Inspection interval" is set to 60 minutes, detection occurs every 60 minutes.

3.7 No Driver

If DMS continuously fails to detect the driver's whole face and this state persists beyond the set sensitivity time duration as shown in Table 1.1, a "No Driver" alarm will be triggered.

Attention: **No Driver** is designed for vehicles in motion. Therefore, an alarm will still be activated if the DMS can only detect part of the driver's face or most of the face, like Figure 13 shows, indicating that the driver is not fully positioned within the cockpit. This partial visibility poses a safety risk, which is why the "No Driver" alarm is triggered. To avoid this situation, please set the sensitivity in "Low" mode or even make it longer in the "User-defined" mode.

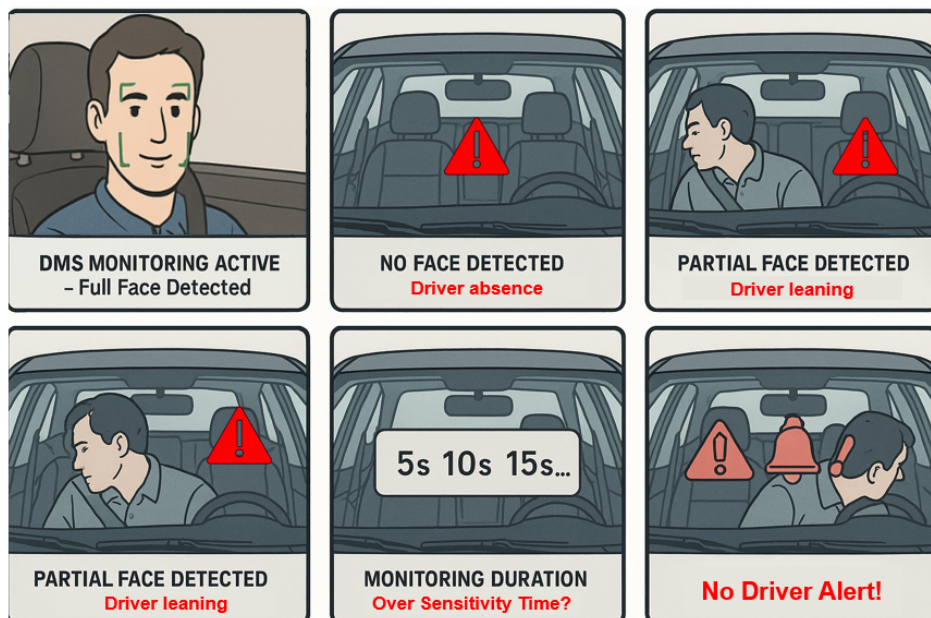


Figure 13: Possible alerts of no driver

3.8 Camera Covering

The DMS performs an analysis of the images. If the image frame is mostly black or dark and this state persists beyond the set sensitivity time threshold, as shown in Table 1.1, a camera cover alarm will be triggered.



Figure 14: Camera cover

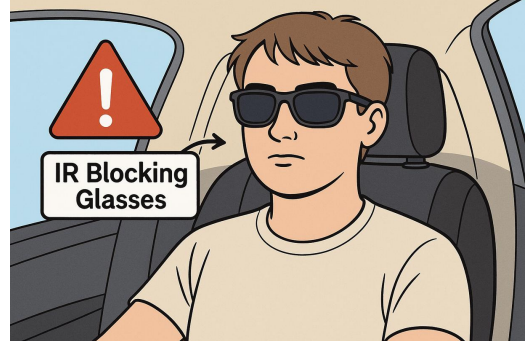


Figure 15: Infrared block alarm

3.9 Infrared block alarm

DMS conducts image recognition feature analysis of the driver's eye area. If infrared-blocking glasses are detected and this state persists beyond the set sensitivity time threshold as listed in Table 1.1, an alarm will be triggered.

3.10 Eat& Drink

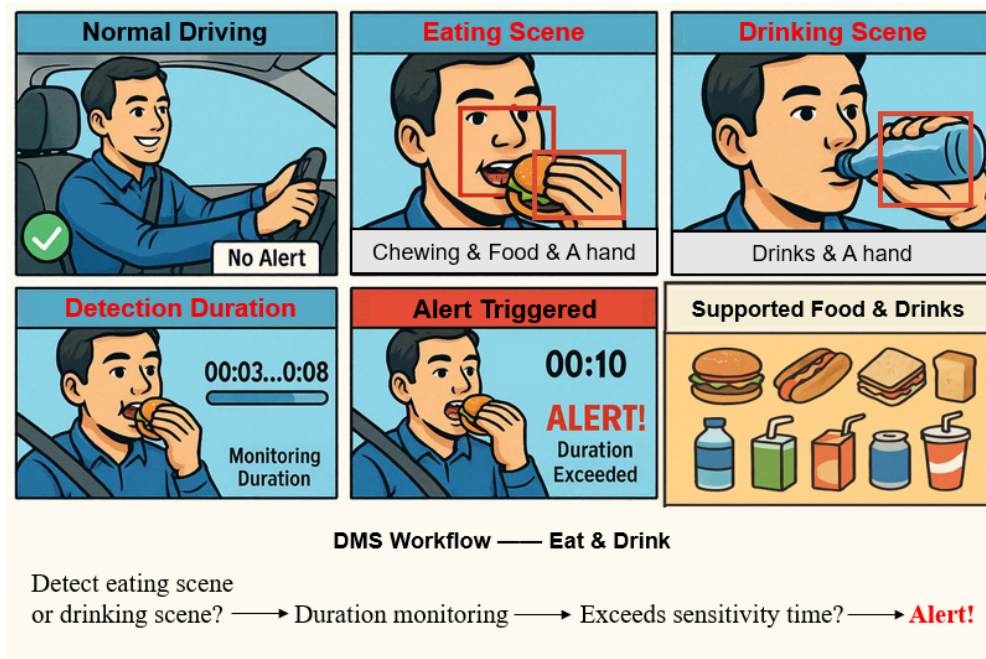


Figure 16: Eat and Drink

The DMS is designed to detect eating or drinking behaviors by the driver, triggering an alarm if these actions persist beyond a configured sensitivity duration as listed in Table

1.1. The detection logic and supported food and drink items are shown in Figure 16.

- **Supported food and drink items:** The system recognizes specific foods, including burgers, hot dogs, sandwiches, and toast. For drinks, it supports 300- 500ml cylindrical bottled beverages, rectangular boxed drinks, cans, and straw-based cola (e.g., McDonald’s-style drinks).
- **Boundary scenarios:** An occluded mouth (e.g., blocked by a hand or object) may cause the algorithm to mistakenly detect eating behavior due to distorted visual data; Drivers wearing gloves may cause hand features to be unrecognised.

3.11 No Mask

When a face is detected, the DMS performs a comprehensive recognition of masks. If a mask is identified as being worn, the system classifies this as a "mask-wearing" state. If this state persists beyond the set sensitivity time threshold as shown in Table 1.1, an alarm will be triggered.

4 ADAS

This chapter details the AI functions of ADAS (Advanced Driver-Assistance System) to provide users with clear and accurate guidance on using these features. The document covers seven core functions: Lane Departure Warning, Forward Collision Warning, Headway Monitoring Warning, Pedestrian Collision Warning, Speed Limit Sign, Stop Sign, and Occlusion Warning. Each function is detailed from multiple dimensions, including recognition logic, trigger conditions, operating speed, and sensitivity settings.

4.1 Lane departure warning

The Lane Departure Warning (LDW) system helps drivers stay within their lane by monitoring lane lines and vehicle position. It uses chromaticity differences in road surface images to distinguish lane lines from the road. Once lane lines are stably recognized, the system detects if the vehicle is deviating from the current lane without using the appropriate turn signal. Virtual alarm boundary lines are set within a specific range on either side of the lane lines. A warning is triggered when the vehicle crosses these boundaries.

- **Trigger Conditions and Sensitivity:**

- Level 1 and Level 2 alarm speed ranges: Specific speed thresholds activate different warning levels (determined by static tests).
 - Sensitivity settings:
 - * High: Alarm triggers when the wheel intersects the lane line.
 - * Medium: Alarm triggers when the wheel exceeds the lane line by approximately 0.1 meters.
 - * Low: Alarm triggers when the wheel exceeds the lane line by approximately 0.3 meters.
 - Alarm valid time: Warnings are issued within a defined time frame, with an error margin of about 0.1 meters in static tests.
- **Note:** The left turn signal suppresses only left lane departure warnings, and the right turn signal suppresses only right lane departure warnings.
 - **Operating Speed:** The system activates at vehicle speeds ≥ 20 km/h.

Installation factors affect the reliability of LDW systems:

- **Installation Variability:** Manual measurements of installation height, vehicle width, side margins, and front-end length can introduce errors, affecting alert sensitivity. Additionally, algorithm self-calibration inaccuracies may further impact LDW sensitivity.

4.1.1 Boundary scenarios

LDW systems may fail or misdetect under certain conditions due to environmental or road-related factors:

- **Adverse Weather Conditions:** Rain, snow, fog, sandstorms, low visibility, or strong backlighting can cause road surface reflections or unclear lane markings. The low contrast between lane lines and the road surface may result in poor lane detection, leading to LDW failure or false detections.
- **Worn or Unclear Lane Markings:** When lane lines are faded or unclear, and the road has water stains, deep tire tracks, cement seams, or other linear marks with significant color contrast, these may be mistakenly identified as lane lines, causing false detections.
- **Complex Lane Markings:** Roads with overlapping new and old lane lines can lead to misdetection.
- **No or Unclear Lane Markings:** If the road lacks lane lines or they are not clearly visible, the LDW function may fail.
- **Improper Camera Installation:** Issues such as incorrect parameter settings, camera misalignment (e.g., too far from the vehicle's center, tilted upward/downward, not level), excessive hood exposure, or interference from dashboard objects can impair performance.

4.2 Forward collision warning

The Forward Collision Warning system enhances safety by detecting potential collisions with vehicles ahead. It recognizes vehicles and lane lines in the current lane, measures the relative distance to the front vehicle, and calculates relative speed based on positional changes. The system computes the time to collision (TTC) by dividing the relative distance by the relative speed. A warning is triggered if the TTC is less than the safe reaction and braking time, which adjusts dynamically based on the host vehicle's speed and the front vehicle's status.

- **Sensitivity:** As a critical safety feature, the system's sensitivity and alarm parameters are fixed and non-adjustable.
- **Operating Speed:** The system activates at vehicle speeds ≥ 8 km/h.

4.2.1 Boundary scenarios

FCW systems may fail or misdetect under certain conditions due to environmental, road-related, or system-related factors:

- **Adverse Weather Conditions:** Rain, snow, fog, sandstorms, low visibility, or strong backlighting can impair imaging of the vehicle ahead, causing the FCW function to fail.
- **Lane-Based Detection Logic:** The system only evaluates vehicles within the detected lane lines. If lane lines are unclear, chaotic, or absent due to environmental or road conditions, the system may fail to reliably identify the host vehicle's lane, potentially mistaking an adjacent lane as the host lane and triggering false alarms based on vehicles in adjacent lanes.

- **Speed Source Limitations:** When using standard GPS for speed data, significant delays may compromise the FCW function. High-frequency inertial GPS, high-precision GPS, pulse, or CAN bus are recommended as speed sources. However, if GPS is used and interference causes inaccurate speed data, the function may fail.
- **Incomplete Vehicle Features:** Scenes where the full vehicle features are not visible (e.g., lateral vehicles where only the body is detected, not the rear) may impair detection, causing the FCW function to fail.
- **Image Quality Issues:** Camera lens contamination from rain, reflections, or severe tail-light overexposure can degrade image quality and affect performance.
- **Large Lens Obstructions:** Significant blockages of the camera lens can impair functionality.
- **Improper Calibration:** Incorrect parameter settings, camera misalignment (e.g., tilted upward/downward, excessive hood exposure, or non-level windshield) can lead to system failure.
- **Curved Roads and Oncoming Vehicles:** On distant curved roads, lane lines are difficult to detect, making it hard to determine if a vehicle is in an adjacent lane. High relative speeds of oncoming vehicles may trigger false alarms.
- **Oncoming or Lateral Vehicles:** The FCW function may fail for vehicles approaching head-on or laterally.
- **Unusual Vehicle Shapes:** Certain rare or unconventional vehicle shapes may result in unstable visual perception, potentially affecting alarm accuracy. These cases can be optimized by collecting relevant video footage.

4.3 Headway monitoring warning

The Headway Monitoring Warning system alerts drivers when the following distance to the vehicle ahead is too short. It uses the same vehicle recognition logic as the Forward Collision Warning system. A warning is triggered when the relative headway divided by the host vehicle speed is less than the set headway monitoring time (adjustable from 0.6s to 4.0s via the host interface) and persists for a set duration (adjustable from 0.1s to 30.0s).

- **Trigger Logic and Sensitivity:**

- With custom sensitivity set to 0.6s and a duration of 2s, warnings are triggered when the headway time is $< 0.6s$ for more than 2s, provided the host vehicle is continuously approaching the front vehicle. No warning is issued if the distance is maintained or increased.
- After a warning, if the headway time is between 0.2s and 0.6s, no further alarm is generated unless the headway time exceeds 1.0s (resetting the warning) and then drops below 0.6s again with an approaching trend.
- If the headway time decreases from 0.6s to below 0.2s, another warning is triggered to indicate immediate danger.

– Standard sensitivity levels:

- * Low: 0.6s
- * Medium: 1.0s
- * High: 1.2s

- **Operating Speed:** The system activates at vehicle speeds ≥ 28 km/h.

4.4 Pedestrian collision warning

The Pedestrian Collision Warning system detects pedestrians and cyclists ahead of the vehicle. It calculates the time to collision (TTC) by dividing the relative distance to the pedestrian or cyclist by the relative speed. A warning, displayed as a large red pedestrian icon, is triggered if the TTC is less than the safe reaction and braking time, which adjusts dynamically based on the host vehicle's speed and the pedestrian/cyclist's status. Although pedestrians and cyclists (including bicycles and motorcycles) have distinct lower-level alarm logic, they are reported and displayed as pedestrians.

- **TTC:** An alarm is triggered when the TTC is below a value, though this is a statistical average and may vary based on speed, lateral and longitudinal distances, lane lines, and drivable area.
- **Sensitivity:** As a critical safety feature, the system's sensitivity and alarm parameters are fixed and non-adjustable.
- **Operating Speed:** The system activates at vehicle speeds ≥ 8 km/h.

Pedestrian/Cyclist Not in Collision Path: If a pedestrian or cyclist is within the current lane but not in the vehicle's driving path (determined by vehicle width, lane lines, and lane-change status), no collision risk is detected, and the PCW alarm will not be triggered.

Dangerous Scenarios Only: The PCW alarm is designed for highly dangerous pedestrian/cyclist scenarios. In scenarios where the vehicle is steadily following a cyclist at a constant speed, the PCW alarm will not be triggered.

4.4.1 Boundary scenarios

PCW systems may fail or misdetect under certain conditions due to environmental, road-related, or system-related factors:

- **Adverse Weather and Lighting Conditions:** Poor lighting at night, rain, fog, strong backlighting, or complex backgrounds can significantly affect pedestrian/cyclist detection. Pedestrians may blend into the background, or significant posture changes or occlusions by other objects may lead to missed detections, causing the PCW function to fail.
- **False Positives from Non-Pedestrian Objects:** Human-shaped signs, roadside mannequins, vehicle decals with human images, or roadside billboards with human figures may be mistakenly identified as pedestrians.
- **Pedestrian Crossing Scenarios:** Due to limitations in the camera's field of view, the PCW function may fail in scenarios where pedestrians cross the road.

- **Close-Range Scenarios:** At close distances, interference from the camera's field of view or the vehicle's hood may result in incomplete imaging of pedestrians/cyclists, impacting detection and ranging accuracy. This significantly reduces alarm accuracy in close-proximity collision scenarios.
- **Medium to High-Speed Scenarios:** At speeds above 30 km/h, detection distance is limited due to lens focal length and hardware constraints (e.g., ADplus wide-angle lenses can detect pedestrians/cyclists up to approximately 25 meters). This may lead to reduced accuracy, missed alarms, or delayed alarms. At high speeds (> 60 km/h), the PCW function is not recommended.
- **Lens Obstruction or Contamination:** Camera lens blockages, dirt, or windshield reflections can impair functionality.
- **Improper Camera Installation:** Issues such as incorrect parameter settings, camera misalignment (e.g., tilted upward/downward, non-level, excessive hood exposure, or dashboard clutter interference) can lead to system failure.
- **Speed Source Limitations:** When using standard GPS for speed data, significant delays may compromise the PCW function. High-frequency inertial GPS, high-precision GPS, pulse, or CAN bus are recommended as speed sources. However, if GPS is used and interference causes inaccurate speed data, the function may fail.

4.5 Speed limit warning

Overspeed Trigger

Detection Effective Time	10	(5 ~ 600)Seconds	
Big Icon Reminder Time	1	(1 ~ 10)Seconds	
Warn Speed	0	(0~20mph) Below Limit Speed	
Alarm Speed	0	(0~20mph) Than Limit Speed	
Duration	0	(0 ~ 15)Seconds	
Effective Time	0	(0 ~ 600)Seconds	?
Do not refresh the alarm timer	☐		?
Discard the Speed Limit	☒		
Orientation Change	32	(0 ~ 90)Degrees	
within	4	(0 ~ 20)Seconds	

Cancel
OK

Figure 17: Overspeed Trigeer

- **Detection Effective Time:** How many seconds the vehicle must be continuously over the warning threshold before the alarm is allowed. Typical useful setting: 10–60 s (avoids short-burst false alarms).
- **Big Icon Reminder Time:** How long the big red overspeed icon stays on screen after the alarm triggers. Typical useful setting: 3–5 s.
- **Warn Speed (Start warning XX km/h before limit):**

Speed-limit value – Warn Speed $\leq$ Current speed $<$ Speed-limit value + Alarm Speed

- **Alarm Speed (Trigger alarm XX km/h over limit):** A complete alarm (beeping + red flashing icon) is triggered **only if both** conditions are met simultaneously:
 1. Current speed $>$ Speed-limit value + **Alarm Speed** threshold (i.e. speed $>$ limit + Alarm Speed)
 2. The above over-speed condition lasts **continuously longer** than the set **Detection Effective Time** (sometimes also called “Duration” or “Continuous Time”)

If the over-speed duration is **shorter** than the Detection Effective Time, the system **downgrades** to a Warn Speed Alert.

- **Duration:** Once triggered, beep/voice alert lasts this many seconds (0 = single beep only). Typical useful setting: 3–5 s.
- **Discard the Speed Limit when turning:** If checked: disables overspeed alerts when turning (orientation, configurable) within xx seconds (configurable).

4.6 Rolling stop

The detection is enabled only when vehicle speed $<$ 50 mph (approx. 80 km/h). At higher speeds, the feature is disabled (STOP signs are rare on highways). This ensures accurate detection of rolling stops in real-world urban driving scenarios while minimizing false positives.

Rolling Stop Trigger			
Detection Distance	10	(3 ~ 20)m	
The Minimum Speed is Greater than	0	(0 ~ 5)mph	
in	5	(1 ~ 10)Seconds	
Effective Time	10	(0 ~ 10)Seconds	?
Do not refresh the alarm timer	☐		?
Cancel		OK	

Figure 18: Configuration of Rolling Stop

5 Different trucking industry customer recommended settings

Based on different trucking industries, recommended settings for DMS are provided in this chapter. However, more accurate settings require enterprises to adjust according to safety and management requirements combined with custom settings.

5.1 Long-Haul Trucking

Long-haul trucking is an important component of the freight industry, typically referring to transportation over a single distance exceeding 500 kilometers. The characteristics of this type of transport include long driving distances, extended durations, and relatively uniform road conditions, with the primary risk stemming from driver fatigue. Long-haul trucking is mainly conducted on highways and national roads, where road conditions are relatively simple but monotonous, easily leading to a decline in driver attention and fatigue accumulation. Transport times typically exceed 8 hours and often involve night driving. The value of goods is usually high, requiring strict adherence to transportation safety standards.

Table 5.1: Long-Haul Trucking Recommended Settings Table

Alert Item	Recommended Sensitivity	Reason
Fatigue	High	Extremely high fatigue risk in long-haul driving requires early fatigue warnings
Distraction	High or Medium	Relatively simple highway environment allows necessary mirror checks and navigation
Yawning	High or Medium	Important supplement to fatigue detection, timely identification of fatigue signs
Handheld Phone	High	Strictly prohibited during long-haul driving
Seatbelt Detection	High	Energy from collisions or sudden braking increases dramatically at high speeds, requiring serious attention
No Driver	High or Medium	DMS only activates with driver face detection
Camera Obstruction	High	Ensures continuous operation of the monitoring system
Smoking	High or Medium	Adjust based on enterprise no-smoking policy
Infrared Blocking	High	Fatigue detection failure should promptly alert the driver
Eat & Drink	High	Adjustable per enterprise needs

5.2 Medium-Short Haul Freight

Medium-short haul freight typically refers to transportation over a single distance between 100-500 kilometers, representing a primary business type in the freight industry. Medium-short haul freight falls between long-haul and urban delivery, presenting certain fatigue risks while requiring navigation across various road types. Transport times typically range from 2-8 hours, potentially involving highways, national roads, urban roads, and other conditions.

5.3 Urban Delivery Services

Urban delivery services include express delivery, food delivery, and supermarket delivery, characterized by short driving distances, frequent stops, and complex road conditions. Urban delivery primarily occurs on city roads, requiring frequent stops, starts, and turns, with a highly variable driving environment. Drivers need to frequently check navigation, find addresses, and communicate with customers. Individual trips are typically short, but the workload is high, requiring multiple deliveries within one day. See Table 5.3.

5.4 Hazardous Materials Transportation

Hazardous materials transportation has extremely high safety requirements, where any non-compliant behavior could lead to severe consequences, necessitating the strictest DMS settings. Hazardous materials transportation involves flammable, explosive, toxic, or corrosive substances, imposing extremely high demands on the driver condition. Any

Table 5.2: Medium-Short Haul Freight Recommended Settings Table

Alert Item	Recommended Sensitivity	Reason
Fatigue	High	Relatively high fatigue risk requires early fatigue warnings
Distraction	Medium	Multi-road condition environment requires balancing road observation and distraction detection
Yawning	Medium	Important supplement to fatigue detection, timely identification of fatigue signs
Handheld Phone	High or Medium	Strictly prohibited to ensure driving safety
Seatbelt	High	Serious attention to seatbelt use under mixed road conditions
No Driver	Medium	DMS only activates with driver face detection
Camera Obstruction	High or Medium	Ensures effective operation of the monitoring system
Smoking Detection	Medium	Adjustable based on enterprise no-smoking policy
Infrared Blocking	High	Fatigue detection failure should promptly alert the driver
Eat & Drink	Medium	Allows moderate water intake to replenish energy, adjustable per management needs

behavior that might compromise safety, including fatigued driving, distracted driving, or smoking, is prohibited during transport. See Table 5.4.

Table 5.3: Urban Delivery Services Recommended Settings Table

Alert Item	Recommended Sensitivity	Reason
Fatigue	Medium	Short individual trips result in relatively lower fatigue risk
Distraction	Medium or Low	Urban driving requires frequent observation of pedestrians, vehicles, and traffic signals
Yawning	Medium	Maintains basic fatigue monitoring
Handheld Phone	Medium or Low	May require use of walkie-talkies or customer calls
Seatbelt	Medium or Low	Frequent entry/exit allows more time for seatbelt use
No Driver	Medium or Low	DMS only activates with driver face detection
Camera Obstruction	Medium	Ensures effective operation of the monitoring system
Smoking	Medium	Adjustable based on enterprise no-smoking policy
Infrared Blocking	High	Eye detection failure should promptly alert the driver
Eat & Drink	Medium	Allows moderate food intake to replenish energy, adjustable per enterprise needs

Table 5.4: Hazardous Materials Transportation Recommended Settings Table

Alert Item	Recommended Sensitivity	Reason
Fatigue	High	Absolutely no fatigue driving allowed in hazardous materials transport
Distraction	High or Medium	Requires road observation while maintaining high attention
Yawning	High	Early detection of fatigue signs
Handheld Phone	High	Strictly prohibited
Seatbelt	High	Seatbelt is a basic safety guarantee
No Driver n	High	Prohibits drivers from being out of position during vehicle operation
Camera Obstruction	High	Must ensure normal operation of the monitoring system
Smoking	High	Strictly prohibited in hazardous materials transport
Infrared Blocking	High	Fatigue detection failure should promptly alert the driver
Eat & Drink	High	Strictly limits eating while driving, allows quick water intake if necessary
Mask	/	Set based on hazardous material type and protective requirements

6 Common Issues and Solutions

1. Why does the DMS keep triggering a "No driver" alert when the driver is seated in the cockpit?

Table 6.1: No driver

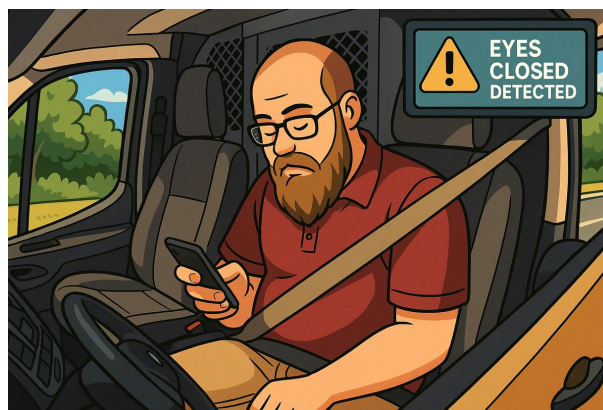
Possible reasons	Recommended solution
Driver leaning against the door, part of face outside cockpit	Adjust sitting posture
Strong lighting prevents clear face detection	Boundary scenarios
DMS cannot see full face (not posture-related)	Reinstall the device
DMS lens obstructed	Remove obstruction

2. Why does the DMS trigger a "No seatbelt" alert when the driver is wearing a seatbelt?

Table 6.2: No seatbelt

Possible reasons	Recommended solution
Steering wheel obstructing the seatbelt	Consider integrating with DSC detection to resolve
Clothing color similar to the seatbelt	Upgrade to the latest algorithm version to resolve
Arm or clothing significantly obstructing the seatbelt	Currently a boundary scenario
Strong light causing overexposure, preventing DMS from detecting the seatbelt	Currently a boundary scenario
Driver clearly wearing the seatbelt, but DMS still triggers an alert	Check if the driver is wearing the seatbelt properly and securely

3. Why does looking down at a phone trigger a fatigue alert by the DMS?



Reason: When the driver glances downward, the imaging of their eyes shows the upper and lower eyelids nearly closed, with the eyeballs not visible. In this scenario, the algorithm may mistakenly detect this as an eye-closed state, triggering a fatigue alert. This issue is a typical algorithm boundary scenario.

Solution: Development has been completed; validation is ongoing.

4. Why does the DMS trigger a "Handheld device" alert when the driver is not using a phone?

Table 6.3: Handheld device

Possible reasons	Recommended solution
Driver holding an object similar in shape to a phone	Some cases are boundary scenarios; the algorithm can effectively filter certain scenarios
Driver holding the steering wheel	Upgrade to the latest algorithm version to resolve
Rear or front passenger using a phone	Currently a boundary scenario

5. What is the difference between Front Collision Warning and Headway Monitoring Warning (HMW)?

Table 6.4: Difference between Front Collision Warning (FCW) and Headway Monitoring Warning

Feature	Front Collision Warning (FCW)	Headway Monitoring Warning (HMW)
Main purpose	Warns of an imminent collision with the vehicle ahead	Warns when you are too close to the front vehicle
What it measures	$TTC = \frac{d_{rel.}}{v_{rel.}}$ $t_h = \frac{d_{rel.}}{v_{host}}$	
When it activates	Only when a crash is likely (usually < 2–3s to impact, often much less)	Much earlier – when time gap drops below threshold

6. How can Camera Cover alerts be configured? Why can't I find it in the "AI APP"?

Click **Config** → **Alarm** → **Video** → **Cover**

you can enable each channel for the corresponding camera that you don't want them to be covered.

The screenshot shows a 'Cover Trigger' configuration window with the following settings:

- Channel:** Four checkboxes for channels 1, 2, 3, and 4.
- Sensitivity:** A dropdown menu set to 'Low'.
- Duration Time:** A text input set to '5' with a range of '(0 ~ 255)Seconds'.
- Effective Time:** A text input set to '10' with a range of '(0 ~ 86399)Seconds' and a question mark icon.
- Do not refresh the alarm timer:** An unchecked checkbox with a question mark icon.
- Speed Range:** A text input set to '≥ 0' with the unit 'KM/H'.
- Buttons:** 'Cancel' and 'OK' buttons at the bottom.

Figure 19: Configuration of Camera Cover